

**Southern Africa
Small Business Markets**

**Local
Shops**

**Informal
Markets**

Financial Health Index for Small Businesses

Predicting Resilience Beyond Profit

Eswatini

Lesotho

Malawi

Zimbabwe

Problem Statement

Many small businesses across Southern Africa are financially fragile — yet invisible to banks, governments, and support systems.



43M+

SMEs excluded from formal credit across Southern Africa



60%

of small businesses fail within 5 years due to financial shocks



0

data-driven holistic tools exist to measure true SME health

Why Profit Isn't Enough

- Banks and governments only focus on profit — a snapshot of today
- Profit doesn't show if businesses can survive crises like illness or drought
- Small businesses are the backbone of the economy in all 4 countries

Profit vs Resilience

Profit (Today)



Shows if you made money now



Snapshot only — one moment



Ignores emergencies

Resilience (Long-Term)



Shows if you can survive crises



Holistic: Savings, Debt, Recovery



Helps plan for Rainy Days

We need BOTH — and machine learning delivers it.

The Solution: Financial Health Index (FHI)

Holistic health check — Goes beyond income to measure real stability

Measures Resilience — Not just profit — actual survival ability

Savings — Safety net available when crises hit?

Debt — Can they manage what they owe?

Recovery — Can they bounce back from a shock?

Financial Health Index (FHI)

01



Savings

*Do they have
a safety net?*

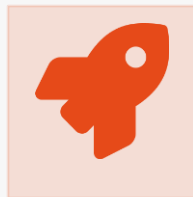
02



Debt

*Can they manage
what they owe?*

03



Recovery

*Can they bounce
back from a crisis?*

What We Want to Achieve



USE MACHINE LEARNING
TO PREDICT BUSINESS
FINANCIAL HEALTH

Using survey data from 4 nations



CATEGORIZE BUSINESSES
INTO: LOW, MEDIUM,
OR HIGH HEALTH

3-class classification output

Output Classifications:

LOW HEALTH

Struggling, No Savings, High Debt

MEDIUM HEALTH

Doing Okay, but Vulnerable

HIGH HEALTH

Strong, Safe, Able to Grow

**Community
Business Owners**

**Informal
Market Traders**

How We Do It

Using Data Clues



Who They Are

Age, gender, education, location, household size



What They Do

Local sales, import/export activity, traded goods



Financial Habits

Banking, mobile money, loans, insurance held



Business Size

Employees vs solo, turnover, expenses, age of firm

**Governments
Target Aid**

**Banks Lend
Responsibly**

**Businesses
Survive & Grow**

Why This Matters



Governments have limited money

They can't help everyone — this tool pinpoints who needs support most.



Banks are afraid to lend

Small shops don't have fancy paperwork — FHI gives them a credible risk score.



It's about the "Rainy Day"

Businesses that survive shocks grow — this model identifies who will make it.

The Machine Learning Approach

01



Collect Data

Survey data from 4 Southern African countries — demographics, finances, attitudes

02



Engineer Features

Create smart indicators like Financial Access Score, Profit Proxy, Insurance Score

03



Train Models

3 algorithms (Random Forest, Gradient Boosting, Extra Trees) vote together as an ensemble

04



Predict Health

Output: LOW, MEDIUM, or HIGH health for every business — with 89% accuracy

Our Results

89%

**Accuracy
(Macro F1 Score)**

Predicting Low / Medium / High
financial health correctly

91%

LOW Health

Struggling businesses correctly flagged for urgent intervention

87%

MEDIUM Health

Vulnerable businesses identified before they fall into crisis

88%

HIGH Health

Resilient businesses recognised for credit and growth support

Tested on real data from Eswatini · Lesotho · Malawi · Zimbabwe

Call to Action / Next Steps

Moving Forward

-  Deploy the FHI model in pilot regions across all 4 countries
-  Collaborate with banks and government agencies on real-world rollout
-  Track improvements in business resilience and financial inclusion over time

Our Vision for Southern Africa



1 Million SMEs

reach financial services they were excluded from



4 Countries

with tools to direct aid exactly where needed



Resilient Economies

built from the ground up, business by business



Global Template

replicable in any developing economy worldwide